

(5/992)

Nepal-Tibet GPS Survey

25 March-12 April 1991

Site Descriptions, Maps and Daily Logs

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Equipment loaned to the project by Ashtech Corporation and Tandy Corporation

543
7-70

खगोल तथा भू-मापन महाशाखा Geodetic Survey Branch

Memo

INDEX

Receiver Oper
Map of Site I
Map showing
Map showing

Location

NEPAL
Bhairhawa
Bharatpur
Biratnagar
Dhankuta
Everest Hotel
Janakpur
Jiri
Jomson
Kalla Patar
Kathmandu
Khadbari
Lukla A
Lukla T
Mahandrenagar
Nagarkot
Nagarkot
Nagarkot
Namche A
Namche B
Namche T
Nepalgarj
Pheriche
Pokhara
Ranja
Simikot
Simira
Surket
Tansen

TIBET

Bala
Dagze
New Tingri
Rongbuk
Xigatse
Daofu
Xialatuo

From: Niraj Mavander
To: Contour

Date:

Time:

- 1) Simikot - 23...
- 2) Mahendranagar - 113...
- 3) Surkhet - 164, 171, 188
- 4) Nepalgarj - 19, 25, 262,
- 5) Ranja - 239.
- 6) Thomsen - 40...
- 7) Pokhara - 67... 92...
- 8) Tansen - 160, 170, 181,
- 9) Bhairawa - 252,
- 10) Bhairatpur - 226,
- 11) Kathmandu - 81...
- 12) Nyeerkat - 1
- 13) Sinola - 109
- 14) Jiri - 103...
- 15) Pheriche - 124... 145,
- 16) Namche - 118... 135, 150
- 17) Lukla - 129...
- 18) Khadbari - 69...
- 19) Hile - 213,
- 20) Tansen - 155, 169, 174
- 21) Biratnagar - 201, 257,

Notes: Bala to Xialatuo near the Xianshuihe fault that were occupied 19-24 April simultaneously with NAGA (08:00-15:20 GMT each day).
XDF2 are two fixed sites near the Xianshuihe Fault will be processed in Beijing.
These data and data from 17 other sites near the Xianshuihe Fault will be processed in Beijing.

Geodetic Library

INDEX

Receiver Operators and affiliations
Map of Site Locations Nepal and Southern Tibet
Map showing Xianshuihe Fault and Tibet
Map showing Xianshuihe Fault Control Points

भूमि पुनरुत्थान तथा भू-संरचना मन्त्रालय
नापी विभाग
खगोल तथा भू-संरचना विभाग
काठमाडौं

Location	Code	Site Description	March 1991					April 1991											
			25	26	27	28	29	1	2	3	4	5	8	9	10	11	12		
NEPAL																			
Bhairhawa	BHAI	207, 252-						208	209	210	211	212							
Bharatpur	BHAR	226						240	241	242	243	244							
Biratnagar	BIRA	201, 257-	202	203	204	205	206												
Dhankuta	HILE	213	221	222	223	224	225												
EverestHotel	VIEW	138									138	139							
Janakpur	JANK	174	155	156	157	158	159												
Jiri	JIRI	103	104	105	106	107	108												
Jomoson	JOMO	40		53	54	55	56	57	58	59	60	61	62	63	64	65	66		
Kalla Patar.	KALA	123													123				
Kathmandu	AIRP	80	87	88	89	90	91												
Khadbari	KHAN	70						75	76	77	78	79							
Lukla A	LUKL	143			27	28	29												
Lukla T	LUKL	140				28	29												
Mahandrenagar	MAHA	113											115	116	117	118	119		
Nagarkot	NAGA	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15		
Nagarkot	NAGD	0			3	4	5	6	7	8	9	10	11	12	13	14	15		
Nagarkot	NA86	14													13				
NamcheA	NAMA	150						135	136										
NamcheB	NAMB	150								137									
NamcheT	NAMC	150						118	119	110	111	112							
Nepalganj	NEPA	195, 262-	18	19	20														
Pheriche	PHER	145											196	197	198	199	200		
Pokhara	POKH	92	67										124	125	126	127	128		
Ranja	RANJ	240											98	99	100	101	102		
Simikot	SIMI	23											248	249	250	251	252		
Simira	SIMR	109						31	32	33	34	35	36	37	38	39			
Surket	SURK	171, 188-						110	111	112									
Tansen	TANS	170, 181-						160	161	162	163	164		164	165	166	167	168	
TIBET																			
Bala	BALA	284-290	315	316	317	318	319												
Dagze	DAGZ	275-283	334	335	336	337	338												
New Tingri	NTGR	298-305						325	326	327	328	329							
Rongbuk	RONG	306-314						330	331	332	332	333							
Xigatse	XIGA	291-292																	
Daofu	XDF2																		
Xialatuo	XDF1																		

Notes: Bala to Xiagatze were occupied with WM102 receivers, Daofu and Xialatuo were occupied with Trimble 4000SST receivers. XDF1 and XDF2 are two fixed sites near the Xianshuihe fault that were occupied 19-24 April simultaneously with NAGA (08:00-15:20 GMT each day). These data and data from 17 other sites near the Xianshuihe Fault will be processed in Beijing.

Nepal 1991 GPS Survey

Nepal GPS Participants

Management

1. Buddhi N. Shrestha, Principal Coordinator
2. Rajendra P. Maratha, Manager
3. Gajanada Adiga, Operations
4. Kiran B. Shrestha, Secretarial Services

Field Team

1. Tirtha B. Pradhananga, Chief Survey Officer
2. Krishna R. Adhikari, Senior Survey Officer
3. Bishnu P. Phuyal, Senior Survey Officer
4. Durgendra M. Kayastha, Survey Officer
5. Basantra Sharma, Survey Officer
6. Karuna B. Shrestha, Survey Officer
7. Gyani R. Chirakar, Seismologist

Reserve

1. Dr. Keshav R. Joshi
2. Narayan B. Karki
3. Amir M. Sakya
4. Maheswor Bhattarai

U.S. GPS Participants

Management

1. Dr. Roger Bilham, Co-principal Investigator
2. Dr. Peter Molnar, Co-principal Investigator
3. Bruce Stephens, Senior UNAVCO Engineer
4. Jim Normandeau, UNAVCO Engineer

Field Team

1. Dr. Peter Molnar, Geophysicist
2. Jim Normandeau, Geodetic Engineer
3. Sergio Barrientos, Geophysicist
4. Jeff Behr, Geophysicist
5. Lin Denham, GPS systems Operator
6. Dr. Darrell Kauffman, Geologist
7. Robert DeConto, Geologist
8. Michael Jackson, Geologist

Reserve

1. Sarah Neustadt
2. Melinda Bell

GPS Systems and Operators

Team	Operator	UNAVCO System	Session 1	Session 2	Session 3
A	Jackson	LA01	Lukla	Namche	Pheriche
B	Molnar	LA02	Simikot	Simikot	Simikot
C	Kaufman	LA03	Jomsom	Jomsom	Jomsom
D	Bilham	UN08	Nagarkot	Nagarkot	Nagarkot
E	Barrientos	UN09	Jiri	Simara	Nepalgunj
F	Normandeau	UN10	Biratnagar	Bhairhawa	Mahendranagar
G	DeConto	LA04	Khadbari	Airport (Kath)	Pokhara
H	Behr	NC01	Dhankuta	Bharatpur	Ranja
I	Denham	NC02	Jankapur	Tansen	Surket

Spare UN07 operated at Nagarkot with Ashtech XII. Second Ashtech XII operated in Khumbu by Stephens

Brief Summary of the Project

A proposal was submitted to NSF by Bilham and Molnar in 1986 for GPS studies of the Himalayan convergence processes in Pakistan and Nepal. Despite approval by the collaborating institutions in each country, government approval was not obtained from either country in 1987. Mike Mayhew at NSF must be commended for maintaining the project status as a funded project for the following 4 years.

The proposed start to observations was delayed until April 1988 and a reconnaissance visit by Ken Hurst and Jim Greer to determine control points to be occupied was made to Nepal in February and March 1988. The field work was postponed for a year in late April when it was clear that government approval would be further delayed. Visits by Bilham to both locations revealed that the defense departments of both countries were blocking permission for GPS studies.

Political developments in Pakistan in 1988 have resulted in this segment of the project being postponed indefinitely. A visit by Shrestha and Tabadar later in 1988 opened the way to further dialog for the Nepal project in 1989. Approval for the Nepal project was not granted until early 1990. A second visit by Shrestha and Tabadar resulted in a signed agreement between the University of Colorado and HMG Nepal in July 1990.

Extensive site reconnaissance and logistic planning was undertaken by Mike Jackson between December 1990 and February 1991. Jackson also visited the PRC in late February to demonstrate operating methods to the Beijing and Xian groups. It is largely due to his efforts that the project was completed successfully in April.

In March 1991 11 GPS operators left for Nepal (see operator listing), and two for Tibet (Paul Bodin and Freysteinn Sigmundsson) to complete the survey in three weeks. A one day equipment check was sufficient in Kathmandu since all operators had trained in early March in Boulder. Each US operator was accompanied to the field by a local geodesist and a driver. Helicopter support was needed in the Khumbu region and this was partly funded by Brad Washburn of the Boston Museum of Science.

We operated a Trimble 4000SST at Nagarkot for an additional ten days after the survey because the second Chinese team operating independently in Eastern Tibet was unable to overlap our main survey dates.

The Faller absolute-g gravimeter was shipped to Kathmandu by NGS (funded by a NASA grant to the University of Colorado). Jack Shied, Brent Bernard and Dan Wineser operated the unit (and two LaCoste Model-D gravimeters) to install an absolute reference point at Nagarkot (NAGA), and two secondary points at Simira (SIMR) and Kathmandu airport (AIRP). The measurements were conducted over a period of ten days during the GPS survey. The NAGA GPS receiver was moved to permit leveling ties and secondary Model-D gravity ties to local points.

A simultaneous survey of northern India, although approved by the Survey of India, was not cleared by the Indian Government in time and is currently scheduled for Dec 1991 completion.

The equipment was shipped across the Pacific and was delayed for several days in Bangkok (total transit 10 days). All the equipment was returned to Boulder via Frankfurt. Freight customs clearance was completed inward in Kathmandu in 2-3 days. Hand baggage (receiver UN07) cleared customs in about 10 minutes with assistance from the Survey of Nepal. All tripods were provided by the Survey of Nepal.

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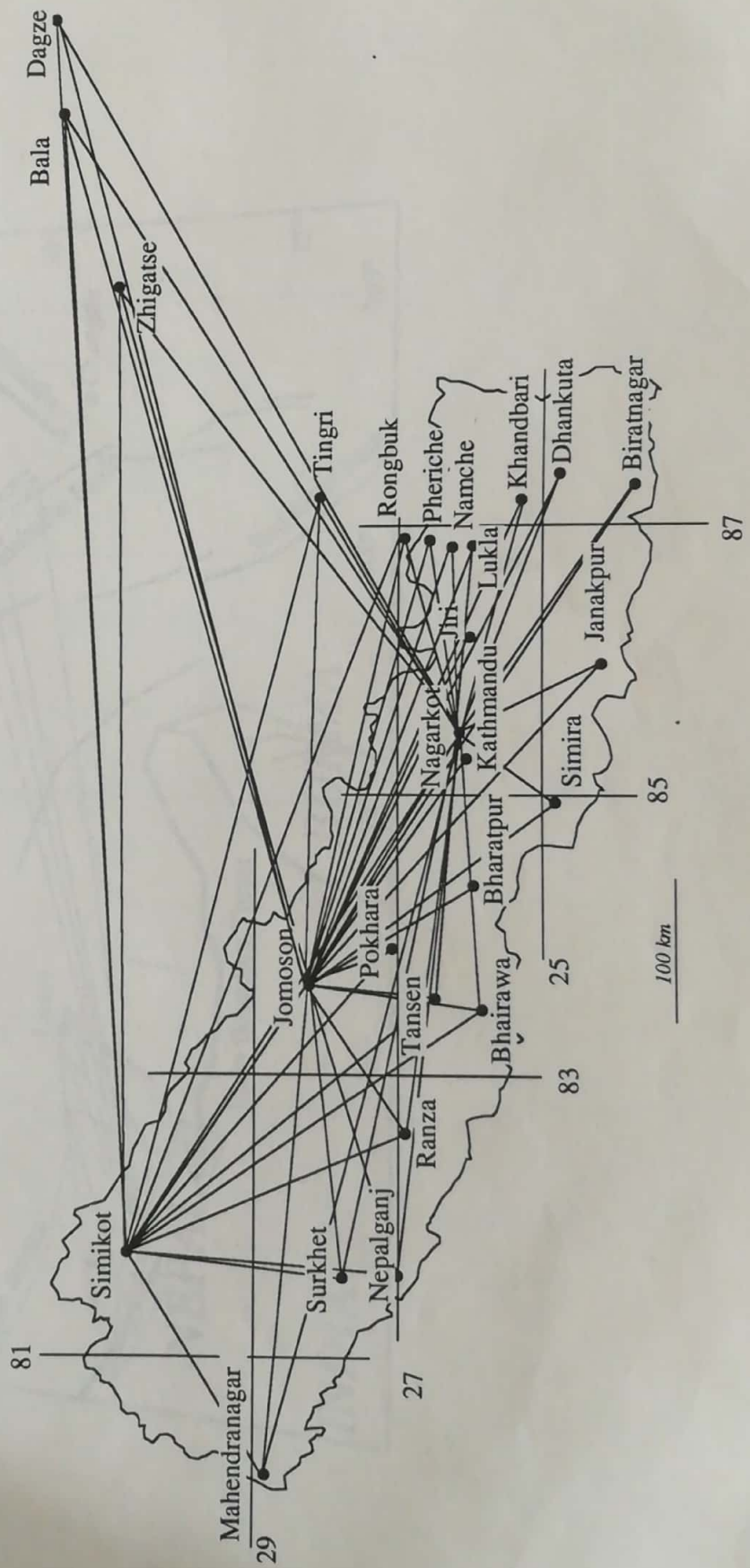
1. Sarah Neustadt
2. Melinda Bell

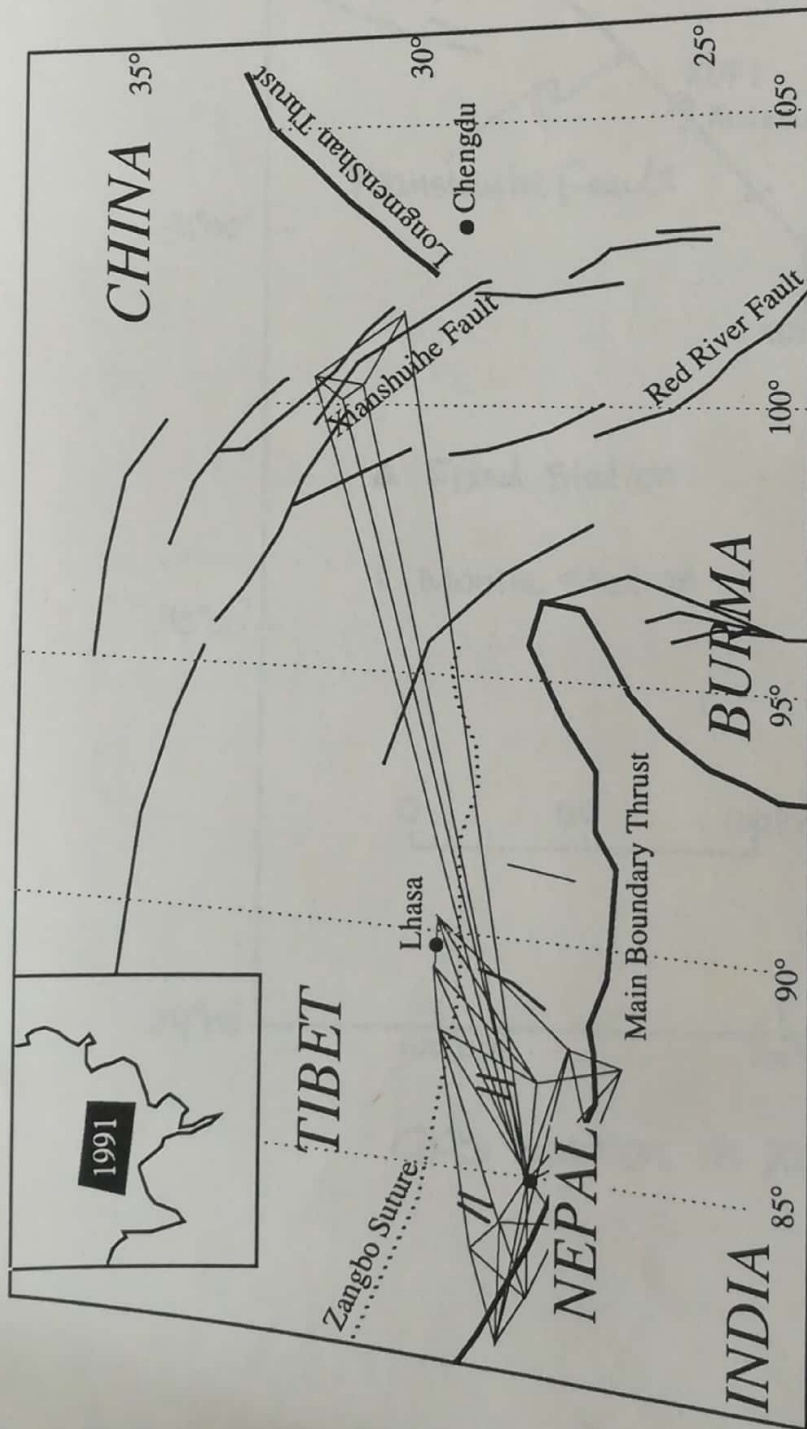
GPS Systems and Operators

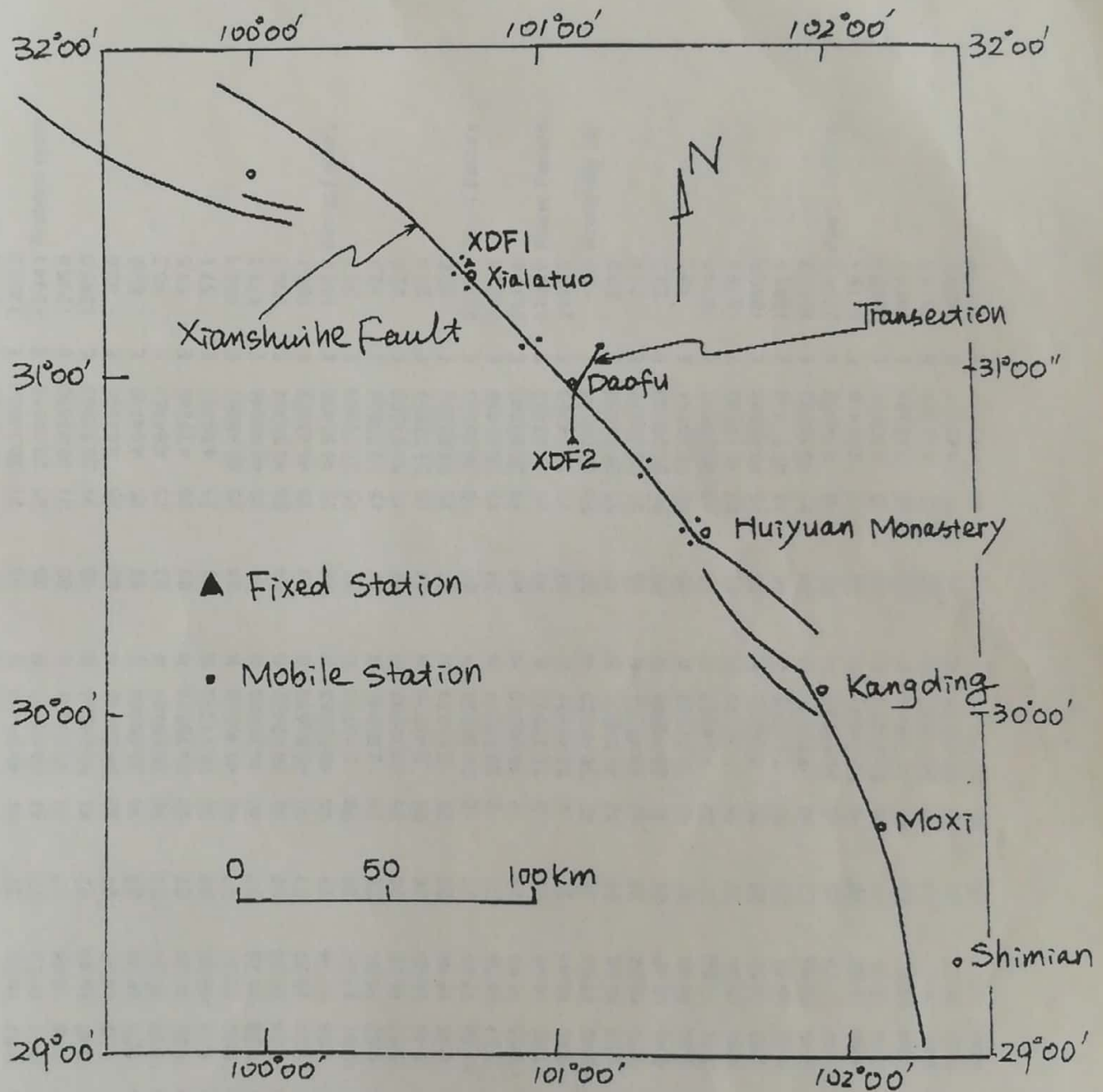
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NEPAL/TIBET GPS Survey
 Direct ties to base stations at Simikot, Jomoson and Nagarkot 25 March-12 April 1991

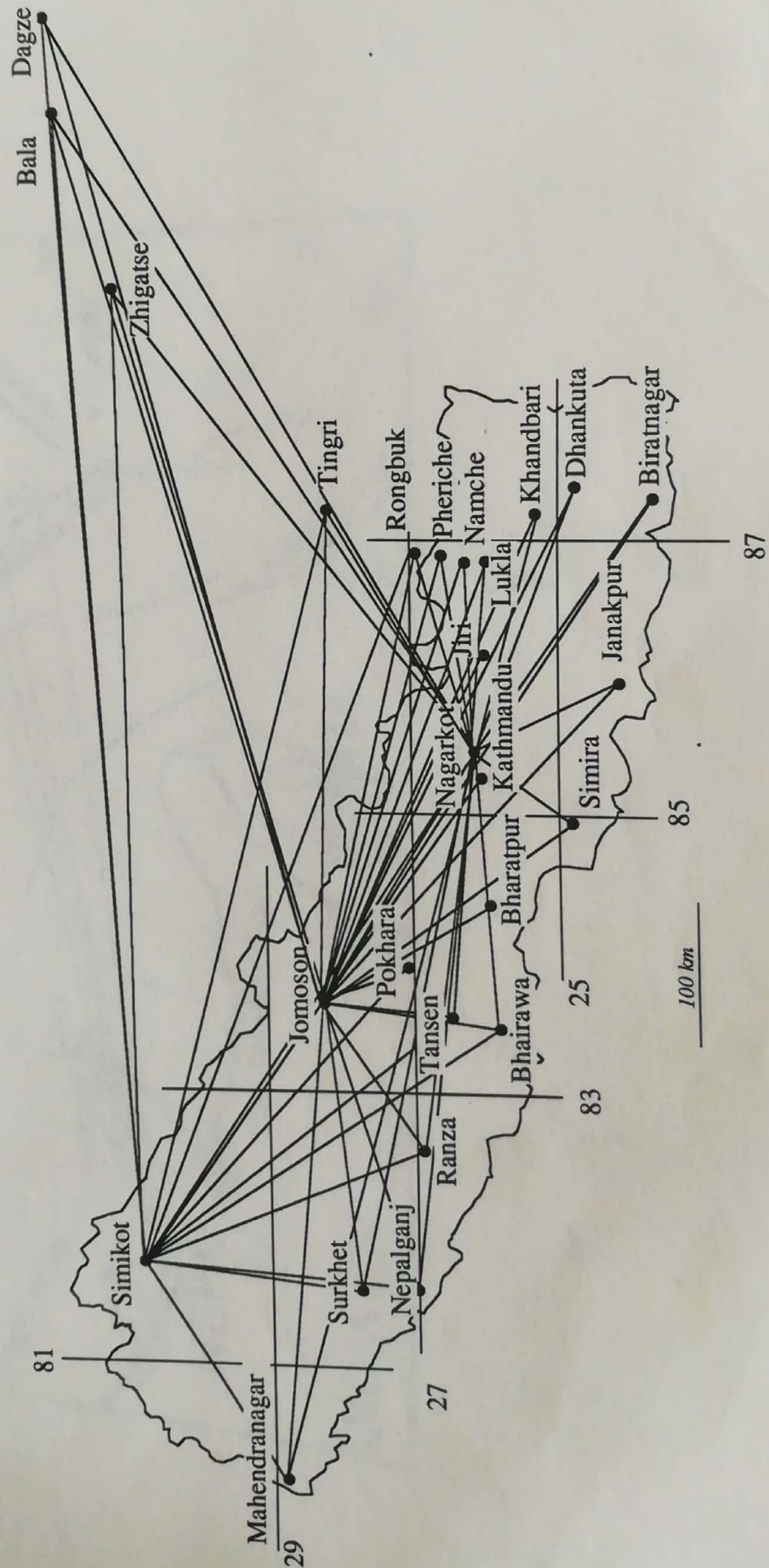






GPS station in Xianshuihe Fault

NEPAL/TIBET GPS Survey Direct ties to base stations at Simikot, Jomoson and Nagarkot 25 March-12 April 1991



Station	Day	Session	Mon.	Day	Sint.Hlt.	Vert.Hlt.	Start	Stop	Deg Lat	Min Lat	Sec Lat	N/S	Deg Lon	Min Lon	Sec Lon	E/W	Height	Problems?
AIRP	84	1	Mar	25	1.4660	1.4536	11:30	17:00	27	41	50.01105	N	85	21	28.43153	E	1304.6	
AIRP	85	1	Mar	26	1.5660	1.5548	9:57	16:56	27	41	49.94466	N	85	21	28.34884	E	1282.2	
AIRP	86	1	Mar	27	1.5290	1.5174	9:53	16:52	27	41	49.88697	N	85	21	28.41615	E	1294.1	Problem readi
AIRP	87	1	Mar	28	1.4650	1.4526	9:48	16:48	27	41	49.87901	N	85	21	28.41650	E	1295.6	
AIRP	88	1	Mar	29	1.5500	1.5386	9:45	16:44	27	41	50.09581	N	85	21	28.53599	E	1295.0	
BHAI	91	0	Apr	1	1.3560	1.3421	9:32	16:32	27	30	26.64946	N	83	25	4.81306	E	45.6	
BHAI	92	0	Apr	2	1.3250	1.3106	9:28	16:28	27	30	26.60464	N	83	25	4.94749	E	62.6	
BHAI	93	0	Apr	3	1.4870	1.4749	9:24	16:24	27	30	26.69107	N	83	25	4.84464	E	48.3	
BHAI	94	0	Apr	4	1.4150	1.4019	9:20	16:20	27	30	26.75529	N	83	25	4.77149	E	57.5	
BHAI	95	0	Apr	5	1.2730	1.2577	9:16	16:16	27	30	26.71389	N	83	25	4.86107	E	57.1	
BHAR	91	0	Apr	1	1.6120	1.6013	9:32	16:32	27	40	40.46508	N	84	25	46.48861	E	159.4	
BHAR	92	0	Apr	2	1.6120	1.6013	9:28	16:28	27	40	40.35008	N	84	25	46.58046	E	175.3	
BHAR	93	0	Apr	3	1.6120	1.6013	9:24	16:24	27	40	40.49432	N	84	25	46.51301	E	159.7	
BHAR	94	0	Apr	4	1.6120	1.6013	9:20	16:20	27	40	40.51639	N	84	25	46.43082	E	168.1	
BHAR	95	0	Apr	5	1.6120	1.6013	9:25	16:16	27	40	40.52563	N	84	25	46.53456	E	169.2	Power Failure
BIRA	84	0	Mar	25	1.5070	1.4951	10:01	17:00	26	29	2.40758	N	87	15	51.79099	E	36.1	
BIRA	85	0	Mar	26	1.4150	1.4019	9:57	16:56	26	29	2.36946	N	87	15	51.73626	E	12.3	
BIRA	86	0	Mar	27	1.3870	1.3735	9:53	16:52	26	29	2.33293	N	87	15	51.84598	E	22.6	
BIRA	87	0	Mar	28	1.5230	1.5113	9:49	16:48	26	29	2.43579	N	87	15	51.77857	E	27.4	
BIRA	88	0	Mar	29	1.4680	1.4556	9:45	16:44	26	29	2.69279	N	87	15	51.88301	E	22.6	
HILE	84	0	Mar	25	1.5730	1.5619	11:40	17:00	27	2	58.30236	N	87	19	36.70533	E	2127.4	Power Failure
HILE	85	0	Mar	26	1.5740	1.5629	9:57	16:56	27	2	58.13204	N	87	19	36.61436	E	2105.6	
HILE	86	0	Mar	27	1.5740	1.5629	9:53	16:52	27	2	58.09860	N	87	19	36.66247	E	2117.6	
HILE	87	0	Mar	28	1.5740	1.5629	13:57	16:48	27	2	58.17517	N	87	19	36.63463	E	2120.6	Power Failure
HILE	88	0	Mar	29	1.5740	1.5629	9:45	16:44	27	2	58.43191	N	87	19	36.74263	E	2118.5	
JANK	84	1	Mar	25	1.3690	1.3553	15:42	17:00	26	42	38.46105	N	85	55	28.04344	E	-2.8	Wrong day ide
JANK	85	0	Mar	26	1.3796	1.3660	9:57	16:56	26	42	38.54032	N	85	55	28.07055	E	22.5	
JANK	86	0	Mar	27	1.3163	1.3017	9:53	16:52	26	42	38.45362	N	85	55	28.22832	E	24.4	
JANK	87	0	Mar	28	1.3295	1.3152	9:48	16:48	26	42	38.44968	N	85	55	28.20050	E	23.3	
JANK	88	0	Mar	29	1.3090	1.2943	9:44	16:44	26	42	38.71922	N	85	55	28.30751	E	18.5	
JIRI	84	0	Mar	25	1.7340	1.7245	10:01	17:00	27	38	7.64900	N	86	13	49.27692	E	1894.0	
JIRI	85	0	Mar	26	1.6730	1.6629	9:50	17:00	27	38	7.78138	N	86	13	49.17620	E	1878.0	
JIRI	86	0	Mar	27	1.7015	1.6917	9:53	16:52	27	38	7.58032	N	86	13	49.16644	E	1882.8	
JIRI	87	0	Mar	28	1.7405	1.7311	9:49	16:48	27	38	7.66555	N	86	13	49.21340	E	1885.5	
JIRI	88	0	Mar	29	1.6495	1.6392	9:45	16:44	27	38	7.82671	N	86	13	49.34792	E	1881.2	
JOMO	85	0	Mar	26	0.8920	0.8672	11:40	17:05	28	46	50.59825	N	83	43	4.14089	E	2799.5	Power failure
JOMO	86	0	Mar	27	0.8910	0.8662	9:46	16:52	28	46	50.70831	N	83	43	4.31418	E	2819.1	
JOMO	87	0	Mar	28	0.8910	0.8662	9:34	16:48	28	46	50.60376	N	83	43	4.30601	E	2807.4	
JOMO	88	0	Mar	29	0.8910	0.8662	9:24	16:44	28	46	50.79004	N	83	43	4.47384	E	2814.1	
JOMO	91	0	Apr	1	0.8910	0.8662	9:16	16:32	28	46	50.79597	N	83	43	4.31265	E	2814.3	
JOMO	92	0	Apr	2	0.8910	0.8662	9:08	16:28	28	46	50.63037	N	83	43	4.43801	E	2823.1	
JOMO	93	0	Apr	3	0.8910	0.8662	9:06	16:24	28	46	50.75679	N	83	43	4.33397	E	2807.8	

Station	Day	Session	Mon.	Day	Sint Hts.	Vert Hts	Start	Stop	Deg Lat	Min Lat	Sec Lat	N/S	Deg Lon	Min Lon	Sec Lon	E/W	Height	Problems?
JOMO	94	0	Apr	4	0.8910	0.8662	9:00	16:20	28	46	50.85625	N	83	43	4.23545	E	2824.5	
JOMO	95	0	Apr	5	0.8910	0.8662	8:56	16:16	28	46	50.77560	N	83	43	4.35264	E	2817.9	
JOMO	98	0	Apr	8	0.8910	0.8662	8:44	16:04	28	46	50.74998	N	83	43	4.53750	E	2805.8	
JOMO	99	0	Apr	9	0.8910	0.8662	8:40	16:00	28	46	50.76303	N	83	43	4.14955	E	2823.8	
JOMO	100	0	Apr	10	0.8910	0.8662	8:36	15:56	28	46	50.75870	N	83	43	4.30682	E	2814.4	
JOMO	101	0	Apr	11	0.8910	0.8662	8:32	15:52	28	46	50.75066	N	83	43	4.22509	E	2812.1	
JOMO	102	0	Apr	12	0.8910	0.8662	8:28	15:48	28	46	50.74335	N	83	43	4.28776	E	2827.7	
KHAN	91	1	Apr	1	1.5420	1.5305	9:33	10:38	27	22	47.08097	N	87	12	19.89763	E	1073.3	Power failure
KHAN	92	1	Apr	2	1.4500	1.4374	9:29	16:28	27	22	47.43767	N	87	12	20.06143	E	1085.9	
KHAN	93	1	Apr	3	1.4080	1.3948	9:25	16:24	27	22	47.46887	N	87	12	19.92584	E	1076.3	
KHAN	94	1	Apr	4	1.6000	1.5892	9:21	16:20	27	22	47.48376	N	87	12	19.89549	E	1075.9	Problem read
KHAN	95	1	Apr	5	1.3920	1.3786	9:17	16:16	27	22	47.55101	N	87	12	19.99819	E	1080.9	
LUKL	86	1	Mar	27	1.4080	1.3948	9:53	16:52	27	41	10.49952	N	86	43	34.35076	E	2750.9	
LUKL	87	1	Mar	28	1.4080	1.3948	10:02	16:48	27	41	10.49702	N	86	43	34.42489	E	2742.2	
LUKL	88	1	Mar	29	1.4080	1.3948	9:54	16:50	27	41	10.67926	N	86	43	34.56011	E	2753.0	
MAHE	98	0	Apr	8	1.7125	1.7028	9:04	16:04	28	57	47.74087	N	80	8	52.72025	E	166.4	
MAHE	99	0	Apr	9	1.6720	1.6619	9:00	16:00	28	57	47.76987	N	80	8	52.48836	E	178.9	
MAHE	100	0	Apr	10	1.6225	1.6119	8:56	15:56	28	57	47.75065	N	80	8	52.46571	E	166.8	
MAHE	101	0	Apr	11	1.6850	1.6751	8:52	15:52	28	57	47.65783	N	80	8	52.36781	E	172.7	
NAGA	84	1	Mar	25	1.2917	1.2767	9:51	12:01	27	41	33.77223	N	85	31	16.17779	E	2133.6	Power fail, re
NAGA	85	1	Mar	26	1.2910	1.2760	9:50	16:40	27	41	33.94589	N	85	31	16.22294	E	2102.4	
NAGA	86	1	Mar	27	1.2912	1.2762	9:40	17:00	27	41	33.83570	N	85	31	16.17689	E	2118.5	
NAGA	87	1	Mar	28	1.2912	1.2762	9:36	16:56	27	41	33.91039	N	85	31	16.27095	E	2119.3	
NAGA	88	1	Mar	29	1.2912	1.2762	9:28	16:48	27	41	34.15746	N	85	31	16.39967	E	2116.7	
NAGA	91	1	Apr	1	1.2912	1.2762	9:23	16:42	27	41	34.01275	N	85	31	16.29155	E	2109.2	
NAGA	92	1	Apr	2	1.2912	1.2762	9:25	16:31	27	41	33.98253	N	85	31	16.39524	E	2124.0	Entered wrong
NAGA	93	1	Apr	3	1.2910	1.2760	9:14	16:34	27	41	34.01991	N	85	31	16.27354	E	2120.5	
NAGA	94	1	Apr	4	1.2910	1.2760	9:11	16:30	27	41	34.08536	N	85	31	16.25298	E	2118.6	
NAGA	95	1	Apr	5	1.2911	1.2761	9:07	16:26	27	41	34.09642	N	85	31	16.35128	E	2121.3	
NAGA	98	1	Apr	8	1.2905	1.2755	8:56	16:14	27	41	34.04706	N	85	31	16.48262	E	2110.0	
NAGA	99	1	Apr	9	1.2905	1.2755	8:50	16:10	27	41	34.06679	N	85	31	16.26810	E	2120.6	
NAGA	100	1	Apr	10	1.2905	1.2755	8:57	16:06	27	41	34.06957	N	85	31	16.35835	E	2109.1	
NAGA	101	1	Apr	11	1.2793	1.2641	8:42	15:42	27	41	33.99297	N	85	31	16.26777	E	2105.2	
NAGA	102	1	Apr	12	1.2800	1.2648	8:38	15:38	27	41	34.01952	N	85	31	16.31058	E	2124.3	
NAGA	108	1	Apr	18	1.2785	1.2633	11:07	15:24	27	41	34.13246	N	85	31	16.53872	E	2100.7	
NAGA	109	1	Apr	19	0.0000		8:20	15:20	27	41	34.04606	N	85	31	16.38336	E	2111.2	
NAGA	110	1	Apr	20	0.0000		8:16	15:16	27	41	34.18843	N	85	31	16.35941	E	2113.1	
NAGA	111	1	Apr	21	0.0000		8:12	15:12	27	41	34.12961	N	85	31	16.46733	E	2095.5	
NAGA	112	1	Apr	22	0.0000		8:08	15:08	27	41	33.99686	N	85	31	16.19732	E	2110.5	
NAGA	113	1	Apr	23	0.0000		8:04	15:04	27	41	34.15641	N	85	31	16.44036	E	2101.6	
NAGA	114	1	Apr	24	0.0000		8:00	15:00	27	41	33.99753	N	85	31	16.35915	E	2116.5	
NAGA	115	1	Apr	25	0.0000		7:56	14:56	27	41	34.02630	N	85	31	16.28537	E	2122.0	

Station	Day	Session	Mon.	Day	Start Ht.	Vert Ht	Start	Stop	Deg Lat	Min Lat	Sec Lat	N/S	Deg Lon	Min Lon	Sec Lon	E/W	Height	Problems?
NAGA	116	1	Apr	26	0.0000		7:52	14:52	27	41	34.14088	N	85	31	16.43385	E	2112.8	
NAMC	91	1	Apr	1	0.8700	0.8444	9:33	16:32	27	48	9.65777	N	86	42	54.50302	E	3523.6	
NAMC	92	1	Apr	2	0.8700	0.8444	9:29	16:28	27	48	9.57530	N	86	42	54.55267	E	3542.4	
NAMC	93	1	Apr	3	0.8700	0.8444	9:25	16:24	27	48	9.69681	N	86	42	54.49027	E	3522.3	
NAMC	94	1	Apr	4	0.8700	0.8444	9:21	16:20	27	48	9.76944	N	86	42	54.44406	E	3533.7	
NAMC	95	1	Apr	5	0.8640	0.8382	9:17	16:16	27	48	9.71394	N	86	42	54.53558	E	3538.8	
NEPA	84	0	Mar	25	1.2700	1.2547	12:45	17:00	28	8	2.92356	N	81	34	28.95551	E	112.8	
NEPA	85	0	Mar	26	1.4793	1.4671	9:56	16:56	28	8	2.95024	N	81	34	28.62378	E	90.1	
NEPA	86	0	Mar	27	1.3808	1.3672	9:52	16:52	28	8	2.84424	N	81	34	28.65458	E	102.4	
NEPA	87	0	Mar	28	1.4662	1.4538	9:48	16:48	28	8	2.84332	N	81	34	28.78835	E	101.9	
NEPA	98	0	Apr	8	1.4870	1.4749	9:04	16:04	28	8	2.98475	N	81	34	28.95501	E	94.8 Antenna Ht no	
NEPA	99	0	Apr	9	1.3830	1.3695	9:00	16:00	28	8	3.00512	N	81	34	28.71248	E	105.2	
NEPA	100	0	Apr	10	1.3980	1.3847	8:56	15:56	28	8	3.00037	N	81	34	28.74359	E	94.3	
NEPA	101	0	Apr	11	1.4790	1.4668	8:52	15:52	28	8	3.00395	N	81	34	28.66439	E	95.0	
NEPA	102	0	Apr	12	1.4840	1.4718	8:48	15:48	28	8	2.98013	N	81	34	28.77475	E	108.0	
PHER	98	1	Apr	8	1.1350	1.1170	9:43	16:04	27	53	37.70771	N	86	49	22.28320	E	4364.5 40 min late st	
PHER	99	1	Apr	9	1.1350	1.1170	8:56	16:00	27	53	37.68233	N	86	49	22.02627	E	4379.0	
PHER	100	1	Apr	10	1.1350	1.1170	8:51	15:56	27	53	37.67721	N	86	49	22.09165	E	4363.7	
PHER	101	1	Apr	11	0.8750	0.8496	8:48	15:52	27	53	37.60423	N	86	49	21.99013	E	4356.0	
PHER	102	1	Apr	12	0.8740	0.8486	8:44	15:48	27	53	37.67142	N	86	49	22.09861	E	4378.5	
POKH	84	0	Mar	25	0.8690	0.8434	10:33	16:23	28	11	56.44496	N	83	58	39.65120	E	790.2 Wrong day ide	
POKH	98	1	Apr	8	1.2550	1.2394	9:04	16:04	28	11	56.45856	N	83	58	39.77958	E	773.2	
POKH	99	1	Apr	9	1.2830	1.2679	9:00	16:00	28	11	56.47955	N	83	58	39.42122	E	781.2	
POKH	100	1	Apr	10	1.3820	1.3684	8:56	15:56	28	11	56.54515	N	83	58	39.63464	E	775.0	
POKH	101	1	Apr	11	1.2980	1.2831	8:52	15:52	28	11	56.38812	N	83	58	39.49817	E	772.4	
POKH	102	1	Apr	12	1.3060	1.2913	8:49	15:48	28	11	56.48035	N	83	58	39.58274	E	787.6	
RANJ	98	0	Apr	8	1.7590	1.7497	9:04	16:04	28	3	45.36370	N	82	34	23.07999	E	1661.3	
RANJ	99	0	Apr	9	1.7470	1.7376	9:00	16:00	28	3	45.37990	N	82	34	22.77047	E	1670.5	
RANJ	100	0	Apr	10	1.7470	1.7376	8:56	15:56	28	3	45.36352	N	82	34	22.87141	E	1664.2	
RANJ	101	0	Apr	11	1.7470	1.7376	11:05	15:52	28	3	45.55303	N	82	34	22.70801	E	1655.9 Power Failure	
RANJ	102	0	Apr	12	1.7470	1.7376	8:48	15:48	28	3	45.35408	N	82	34	22.86098	E	1675.7 Antenna Ht. re	
SIMA	91	0	Apr	1	1.6405	1.6301	9:32	16:32	27	9	45.05059	N	84	58	54.16280	E	72.7 Entered wrong	
SIMA	92	0	Apr	2	1.6870	1.6771	9:28	16:28	27	9	45.02527	N	84	58	54.26955	E	92.5 Entered wrong	
SIMA	93	0	Apr	3	1.7055	1.6958	9:24	16:24	27	9	45.05493	N	84	58	54.12143	E	79.5 Entered wrong	
SIMI	92	0	Apr	2	1.2427	1.2269	9:28	16:28	29	58	1.20995	N	81	49	35.36339	E	2926.3	
SIMI	94	0	Apr	4	1.2432	1.2274	9:20	16:20	29	58	1.55434	N	81	49	35.22421	E	2925.9	
SIMI	95	0	Apr	5	1.2430	1.2272	9:16	16:16	29	58	1.34734	N	81	49	35.28852	E	2923.5	
SIMI	98	0	Apr	8	1.2432	1.2274	9:04	16:04	29	58	1.33302	N	81	49	35.39966	E	2910.2	
SIMI	99	0	Apr	9	1.2430	1.2272	9:03	16:00	29	58	1.42171	N	81	49	35.26612	E	2927.7	
SIMI	100	0	Apr	10	1.2432	1.2274	8:56	15:56	29	58	1.49222	N	81	49	35.31456	E	2913.9	
SIMI	101	0	Apr	11	1.2427	1.2269	8:53	15:52	29	58	1.50949	N	81	49	35.14377	E	2913.0	
SURK	98	0	Apr	8	1.4557	1.4432	9:04	16:04	28	35	8.95704	N	81	38	6.90832	E	635.0	

Station	Day	Session	Mon.	Day	Sht Ill.	Vert Ill.	Start	Stop	Deg Lat	Min Lat	Sec Lat	N/S	Deg Lon	Min Lon	Sec Lon	E/W	Height	Problems?
SURK	99	0	Apr	9	1.4673	1.4549	9:00	16:00	28	35	9.00052	N	81	38	6.65263	E	644.8	
SURK	100	0	Apr	10	1.4923	1.4802	8:56	15:56	28	35	8.98605	N	81	38	6.70189	E	636.2	
SURK	101	0	Apr	11	1.5402	1.5287	8:52	15:52	28	35	8.94480	N	81	38	6.58090	E	637.8	
SURK	102	0	Apr	12	1.4845	1.4723	8:48	15:48	28	35	8.95640	N	81	38	6.70285	E	650.1	
TANS	91	0	Apr	1	1.4222	1.4092	9:32	16:32	27	52	26.05179	N	83	33	13.63918	E	1397.6	Power Failure
TANS	92	0	Apr	2	1.3415	1.3273	9:28	16:28	27	52	25.96069	N	83	33	13.78369	E	1415.2	
TANS	93	0	Apr	3	1.4278	1.4149	9:24	16:24	27	52	26.08492	N	83	33	13.69384	E	1399.1	
TANS	94	0	Apr	4	1.4953	1.4833	9:20	16:20	27	52	26.12234	N	83	33	13.56840	E	1409.0	
TANS	95	0	Apr	5	1.4213	1.4083	9:17	16:16	27	52	26.10260	N	83	33	13.69993	E	1407.5	